The mass gap: current position of this programme

Later corrections: the critical-coupling note withdraws the phase-transition equivalences of T2′; the transfer note withdraws the verbatim transfer of Euclidean estimates; the action lower-bound note and the entropy note withdraw sharpness and the entropy reading of the mode count; the instability note and the Jacobian note withdraw sharp growth and the Hamiltonian truncation error; the Kogut–Susskind note withdraws its explicit threshold; the openings note §1 corrects the all-coupling requirement; the October 1 audit §3 lists this note's inconsistent numbers (relevant passages read).

> **Correction (2026-10-02).** This map kept several claims after their source notes corrected them, and it quoted thresholds that did not form one labelled set. The changes, made in place below:
>
> 1. *Phase reading of T2′.* The identification of T2′ with absence of a zero-temperature bulk transition, and on a second-order branch with uniqueness of the continuum limit, is **withdrawn**, following the critical-coupling note. The statement that T2′ fails for $SU(N \geq 5)$ is **withdrawn**: the first-order bulk transition there is numerical evidence for the Wilson action, a different operator, and the spectral comparison with the Kogut–Susskind gap is missing. "False for $U(1)$" is downgraded likewise: the massless phase is proved for Euclidean actions (Guth for the Villain action; Fröhlich–Spencer) and is not transferred to this operator here.
> 2. *Large field.* "The large-field region closed in measure" is **withdrawn**. What survives is a continuum action floor $\eta_F^2/(4g^2)$ for a flowed block, under flow and boundary assumptions, with sharpness open; and a phase-space count $\eta_b^2/(8\pi^2)$ for one unit-charge complex field in a constant background, a model that bounds no entropy. The two η are different parameters. The verbatim import of Euclidean estimates into the Kogut–Susskind ground state is **withdrawn**; the slice identity stands. "The flow's growth cannot be improved" is withdrawn with the instability note's sharpness claim.
> 3. *Truncation.* The Hamiltonian truncation error Ce^{-cK} at range Ka (§2, §3, §6) is downgraded to a target: the corrected Jacobian note keeps kernel-tail budgets and leaves their conversion into a Hamiltonian error open.
> 4. *Thresholds.* For the $SU(3)$ Wilson transfer matrix the rigorous thresholds are $g^2 > 444$ (Dobrushin) and $g^2 \geq 1059$ (polymer expansion, all representations; 1059 without the source's linearization, §6); 176 is the leading-activity figure only. Their gap rates are now printed exactly, and the former large-g^2 forms, which slightly overstate them, are marked $\simeq$. For the Kogut–Susskind Hamiltonian the volume-uniform gap rests on Yarotsky's existential threshold; a repository proof-extraction estimate of g_0^2 of order 10^{102} is heuristic; the explicit $g^2 \geq 388$ is withdrawn. The weak side carries two labelled estimates: a necessary window $1/g^2 \gtrsim 10^2$ with crude constants, and a one-step remainder threshold $g^2 \sim 10^{-12}$ that assumes an unproved order-one propagator decay rate ($\sim 10^{-35}$ at the proved rate).
> 5. *Scope.* "Exhaust" (§3), "every inequality derived from" the eigenvalue equation (§3) and "any proof of T2′ must use one of them" (§4) are restricted to the routes, estimates and structures examined here. The §5 row that equated T2′ with closedness of the gapped set now states that T2′ needs openness as well.
> 6. *Smaller precisions.* a_* is 0.1 to 0.17 fm, as in the conditional theorem; the range of certified polymer enumeration is the proved $\beta_W \lesssim 10^{-2}$, replacing an unsourced "β_W of order one"; the valley potential is labelled a formal one-loop

calculation; §6 adds the doubling count for the 10^{-12} weak boundary and a §8 states the consequence for STATE.

This is the synthesis of the notes written since the goal was set to a proof of the Yang–Mills existence and mass-gap conjecture. It states what has been proved here, what has been imported, what has been ruled out, and what remains, with constants explicit and each claim labelled.

The short version: the conjecture decomposes into six named statements, of which two are proved here and one is imported. The clause $m < \infty$ is closed as far as variational methods reach, and reduced to the existence of the theory plus the nontriviality of one flowed correlator. The clause $m > 0$ is untouched, and **seven routes to it have been closed by explicit computation**, in two families: those built from operator norms and spectra, and those built from the ground state of the Hamiltonian. Each route fails for a reason stated with explicit constants, and the second family's failure identifies what a factorized Euclidean weight supplies that estimates drawn from the ground-state eigenvalue equation do not. The strong-coupling thresholds are rigorous and explicit for the Wilson transfer matrix; the weak-side thresholds are labelled estimates; the large-field region remains open (§§5–6).

1. The decomposition

The obligations map turns "prove the mass gap" into six statements about the Kogut–Susskind Hamiltonian

$$H = \frac{\hbar c}{a}\Big[\frac{g^2}{2}\sum_{\ell}(-\Delta_\ell) + \frac{2}{g^2}\sum_{p}\big(N - \operatorname{Re}\operatorname{tr} U_p\big)\Big], \qquad \Delta_{a,L} = \frac{\hbar c}{a}\,\delta(g; N_s, G).$$

	statement	status
T1	finite lattice: unique physical ground state, $\delta > 0$	**proved here**
T2	$\delta \geq \gamma(g^2/2)C_2$ for $g \geq g_0$, uniform in volume	**proved here** from Yarotsky's theorem, with an existential g_0; following his proof with numbers gives $g_0^2 \simeq 10^{101}$ for $SU(3)$. The Wilson transfer matrix, a different operator, has explicit rigorous thresholds (§6)
T2′	$\liminf_{N_s} \delta > 0$ for every $g > 0$	open. $U(1)$: a massless phase is proved for Euclidean actions (Guth, Villain action; Fröhlich–Spencer), not transferred to this operator. $SU(N \geq 5)$: numerical evidence of a first-order bulk transition for the Wilson action, a different operator
T3	$\delta_\infty(g)/(a\Lambda_{\mathrm{lat}}(g)) \to m/(\hbar c\Lambda) \in (0, \infty)$	open
T4	continuum infinite-volume theory with the axioms	finite-volume ultraviolet stability known

	statement	status
S	small volume: $\Delta = \delta_1 g^{2/3}\hbar c/L\,[1 + O(g^{2/3})]$	upper side proved, lower side fixed-lattice only

The conjecture is T3, given T4, with a gap $\delta_\infty > 0$ on an interval $(0, g_1)$ along the scaling curve. T2′, a gap of this lattice operator at every coupling, is stronger, and the conjecture does not need it (precision of 2026-09-23; the openings note §1 also records four ways in). For $SU(N \geq 5)$ the Wilson action shows a first-order bulk transition at intermediate coupling in numerical studies. That proves no hole in the gapped set of the Kogut–Susskind operator: the operators differ and the spectral comparison is missing (critical-coupling note §4b, correction of 2026-10-02). What the critical-coupling note proves is that the finite-volume gap is continuous in g, and that openness of the gapped set together with its closedness would give T2′ by connectedness. Its former identification of T2′ with the absence of a zero-temperature bulk transition, and with uniqueness of the continuum limit, is withdrawn.

2. What is proved here

T1 (obligations map §2): compact resolvent, a simple strictly positive ground state, a positive gap inherited by the physical sector, for every a, N_s, g and compact connected G.

T2 in Hamiltonian form (T2 note): the electric term is classical in the Peter–Weyl partition and the magnetic term is a bounded range-$\{0,1\}^3$ perturbation with $\beta = 48N^2/[g^4(N^2-1)]$, so Yarotsky's theorem gives $\Delta_{a,L} \geq \gamma(g^2/2)C_2\hbar c/a$ for $g \geq g_0$, uniformly in the lattice size, with exponential clustering. The threshold g_0 is existential; a conditional proof-extraction estimate suggests an order of 10^{102} for g_0^2 in $SU(3)$ but is not a certified threshold (threshold note), and the explicit threshold $g^2 \geq 388$ of the continuous-time note was withdrawn on 2026-10-02.

A Lieb–Robinson bound (note): velocity $v \leq 32eNc/g^2$, uniform in volume, because the unbounded part of H is a sum of commuting single-link Laplacians; hence a strong-coupling correlation length $\xi \leq C'aN/(\gamma C_2 g^4)$.

Upper bounds (note, Polyakov note): the Feynman–Bijl inequality with explicit constants; the strong-coupling gap pinned to order $g^2\hbar c/a$ on both sides, the lower side for $g \geq g_0$; the abelian gap bounded by the plaquette-sine structure factor; the proof that **no operator linear in the electric field with c-number coefficients is gauge invariant for a non-abelian group**; and a necessary condition on the Polyakov-loop susceptibility.

The finiteness clause reduced (moment hierarchy): $m \leq M_{k+1}/M_k$ for every k, with the ratios decreasing, so the f-sum rule is the weakest member and the next one, $m \leq \hbar C_0(0)/\int_0^\infty C_0$, needs only T4 and $C_0 \not\equiv 0$. The free-field value of the weakest member is $4.26\,\hbar c/\sqrt{8t}$, computed in closed form and confirmed by an a second evaluation of the same sum rule (free-field calculation).

Exact structural identities (note): $\Delta V = 4C_2(N|\mathcal{P}| - V)$, so the magnetic energy shifted by its Haar mean is a Laplacian eigenfunction; $|\nabla V|^2 \leq 16V$ for $SU(2)$; and an exact ground-state sum rule tying $\mathrm{Var}_\Omega(V)$ to kinetic quantities.

Flow facts (conjugation, one-step locality): the Wilson flow is a diffeomorphism of $G^{\mathcal{E}}$, unitarily implemented, so conjugation preserves the spectrum; the lattice flow carries no coupling and a doubling is dimensionless flow time 1/2, so the coefficients of its linearization within one step are bounded by pure numbers. The lattice note concluded that truncating the conjugated Hamiltonian at range Ka costs Ce^{-cK} with pure constants. The corrected Jacobian note (2026-10-02) keeps only kernel-tail budgets and leaves their conversion into a Hamiltonian truncation error open, so that conclusion is a target here.

3. Seven closed routes

route	why it closes
variational upper bounds, for $m > 0$	a spectral measure with all negative moments finite can have $m = 0$
expansion around the free theory	the bound reaches m only at the confinement scale, where its coefficient is nonperturbative
blocking by projection	a surface-to-gap estimate worsens by $3M^2/8$ in the stated strong-coupling asymptotic; all-coupling monotonicity is not proved
conjugation by the flow	the spectrum is invariant; a changed local decomposition requires control of generated terms
flow before decimation	sup norms are bijection-invariant and spectra conjugation-invariant
Agmon estimate on the ground state	$\|\nabla V\|_\infty$ is extensive, so it bounds the global excess only
pointwise Gibbs domination	the comparison function bites only above the extensive threshold $V_* \simeq N\|\mathcal{P}\|$

The first five are the routes built from operator norms and spectra that this series tried; the last two are the two standard ways it tried of extracting a local statement from the ground-state eigenvalue equation. Neither list is shown to be exhaustive. Each closure is a computation with explicit constants.

The reason the last two fail, stated once. Both estimates are derived from $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$ and compare the total potential with the total energy; e_0 is extensive, so they separate configurations only by their global excess (global/local note). The Euclidean weight $e^{-S_w} = \prod_p e^{-(2/g_E^2)V_p}$ factorizes over plaquettes, so a local excess costs a local factor of the unnormalized weight. That is the structural advantage of a factorized weight over these two estimates. It falls short of local probability control, because plaquettes share links and the normalization involves all of them (transfer note §2). The Hamiltonian formulation has a configuration density too, $|\Omega|^2$, with an exact slice identity (transfer note §1); what is missing is a local large-field estimate for it, and measure estimates do not by themselves give the signed operator comparisons a Hamiltonian chain consumes (Jacobian note, correction of 2026-10-02).

4. Where the non-abelian structure has been isolated

Three statements distinguish the groups. They are the places this series has isolated where a proof of T2′ could use structure that the abelian theory lacks. They have distinct scopes, and none is by itself a proof of a non-abelian gap (critical-coupling note §4d).

1. *The commutator potential* of the zero-momentum sector is identically zero for an abelian group; where present, the transverse zero-point energy confines the valley and produces the gap $\delta_1 \hbar^{4/3} g^{2/3} m^{-2/3}$ (G07, claim C133).
2. *Gauge invariance blocks the photon channel*: no c-number-coefficient operator linear in E commutes with Gauss's law for a non-abelian group, so the excitation through which the abelian gap closes must carry a Wilson line (upper-bound note Proposition 6).
3. *The one-loop valley potential* $U(a) = \frac{2\hbar c}{\pi^2 L}\Phi(La)$ vanishes identically in the abelian theory, so the holonomy is free; this is a formal one-loop calculation (valley note §3b).

5. What remains

obligation	kind
a gap on $(0, g_1)$ along the scaling curve	the conjecture's lattice half; T2′ would suffice for positivity, and T2′ follows from openness together with closedness of the gapped set
openness of the gapped set	unproved globally: needs volume-uniform gap stability; the stability route cited in the critical-coupling note §4e has hypotheses not verified for the Kogut–Susskind Hamiltonian, including frustration-freeness
T3	the conjecture, given that gap
T4	construction
local large-field control	open. The slice identity is exact: $\langle \Omega, F\Omega \rangle$ is the long-time limit of normalized heat-semigroup expectations of the Kogut–Susskind Hamiltonian, and the squared Perron vector of a transfer matrix at fixed temporal spacing is the slice marginal of its Euclidean measure (transfer note §1); the Wilson-to-Kogut–Susskind limit needs ground-state projector convergence, and no Euclidean large-field estimate transfers verbatim
the large-field region itself	open; ingredients and obstruction in §6 (an action floor with sharpness open, a model mode count that bounds no entropy, and a crude Hamiltonian operator inequality that loses a binding negative term)

The pattern across these notes is uniform: **statements uniform in the cutoff are renormalization statements, and statements local in space need a local weight.** The strong-coupling theorem is uniform in the volume at fixed cutoff; the small-volume theorem is fixed-lattice; the flowed bounds are finite at fixed lattice and need the flow's renormalization in the continuum; the two ground-state estimates of §3 are global because the eigenvalue equation they start from carries the extensive e_0.

6. Assessment

The route that survives is the constructive one: conjugate by the flow, which is exact; truncate, where the kernel tails are cheap and their conversion into a Hamiltonian error is open (Jacobian note, correction of 2026-10-02); decimate, which is the problem; and iterate about twenty-one times for $SU(3)$ from $g^2_{\mathrm{UV}} = 1/2$ (SU(3) constants §3). The transfer note proves the slice identity for $|\Omega|^2$; its former claim that the identity imports every local Euclidean estimate is withdrawn, since the Wilson-to-Kogut–Susskind limit needs ground-state projector convergence and an isotropic estimate is not a bound for the anisotropic family.

The large-field region is open. Its ingredients are two. A continuum action floor $\eta_F^2/(4g^2)$ holds for a flowed block whose L^2 field norm at flow time $\ell^2/8$ is at least η_F, under flow and boundary assumptions, with sharpness open (lower-bound note). A phase-space count $\eta_b^2/(8\pi^2)$, with $\eta_b = gB\ell^2$, holds for the unstable modes of one unit-charge complex field in a constant chromomagnetic background; it is a model that bounds no entropy (entropy note). The two η are different parameters, so their ratio is no action/entropy balance. In that background the Nielsen–Olesen mode grows at rate at most gB under the linearized flow, and sharpness of the

truncation bound is unproved (instability note). Normalized rarity, polymer convergence and closure of the large-field corner are not proved. What the Hamiltonian chain of lower bounds consumes from that region is an operator inequality, whose proof in the operator-inequality note §4 loses a negative term $C_V|\partial N|\hbar c/a$ on the small-field side, of exactly the kind that binds. A Euclidean polymer expansion would consume a measure statement instead, the suppression $e^{-c\eta^2/g^2}$, which is a heuristic target here (transfer note, lead). That is the computed reason for the Euclidean choice in this series; it does not show that every Hamiltonian route fails.

The coupling map for $SU(3)$. Three regions, with numbers:

region	statement	source
strong, Wilson transfer matrix: $g^2 > 444$ (Dobrushin) or $g^2 \geq 1059$ (polymer, all representations), both rigorous; 176 is the leading-activity figure only	$\Delta_W \geq (\hbar c/a)\log[1/(18(e^{24/g^2}-1))] \simeq (\hbar c/a)\log(g^2/432)$, resp. $\Delta_W \geq (\hbar c/a)\,4\log[1/(176(e^{6/g^2}-1))] \simeq (\hbar c/a)\,4\log(g^2/1056)$, uniform in volume	Dobrushin note, polymer note
strong, Kogut–Susskind: explicit threshold withdrawn 2026-10-02 (formerly $g^2 \geq 388$, 79 adjacent)	finite-volume bound $\frac{8}{3}g^2 - 12\lvert P\rvert/g^2$ only; the volume-uniform KS gap rests on Yarotsky's existential threshold	continuous-time note
weak: no coupling at which a step is proved with explicit constants	necessary window $C_1 g \leq \eta \leq C_2$ for the Hamiltonian large-field inequality, nonempty only for $1/g^2 \gtrsim 10^2$ (crude constants); one explicit small-field blocking step has remainder threshold $g^2 \sim 10^{-12}$ if the propagator decay rate is of order one, which is unproved, and $g^2 \sim 10^{-35}$ at the proved rate (scaling estimates)	operator-inequality note §4, part 1b
intermediate, from the weak side's reach ($g^2 \sim 10^{-2}$ at best, on the crude window) to $g^2 = 444$	no expansion applies; a finite-volume mixing condition (Dobrushin–Shlosman) would give the gap coupling by coupling	finite-verification note

The strong boundary is rigorous and explicit; the weak one is an estimate. The polymer note's $1056 = 6\times176$ linearizes $e^{6/g^2}-1 \simeq 6/g^2$; solving its criterion $e^{6/g^2}-1 \leq 21/(500e^2)$ gives the exact sufficient KP condition $g^2 \geq 6/\log(1+21/(500e^2))$. The sufficient integer bound 1059 also makes the displayed rate with 176 positive; that rate itself requires $g^2 > 6/\log(177/176)$. **Precision, 2026-10-02:** the rounded KP value 1058.6, the integer 1059 and the linearization 1056 are different statements, as distinguished in the maintained polymer note.

The width of the intermediate region in one-loop doublings, $n = (1/g^2_{\text{weak}} - 1/g^2_{\text{strong}})/0.0966$ with $2b_0\log 2 = 0.0966$ for $SU(3)$, is set almost entirely by the weak side:

weak boundary	n
$1/g^2 = 10^2$ (crude window)	$\simeq 10^3$
$g^2 = 10^{-12}$ (one-step scaling estimate, order-one rate assumed)	$\simeq 10^{13}$
$g^2 = 1/2$ (if a small-field expansion reached it)	$\simeq 21$

The first and last rows are from the continuous-time note §5; the middle row is the same formula. The Kogut–Susskind threshold from Yarotsky's theorem, made explicit, is $g_0^2 \sim 10^{101}$ and plays no role in the map.

The map as a theorem. The conditional theorem states the whole map as two hypotheses, control of the blocking steps from the weak side to the coupling where $\xi \simeq a$ (H1) and certified mixing at that one coupling on one box (H2), with the conclusion $m \geq \hbar c \min(\gamma_*, \gamma')/a_*$, where a_* is the physical spacing at the verified coupling, about 0.1 to 0.17 fm by the published scale. Progress is progress on H1 or H2.

The two reasons to stop, researched. A Griffiths-type inequality would transfer control from weaker to stronger coupling only and would replace the strong-side steps, never the weak side (research note); a certified verification at $\beta_W \simeq 6$ is beyond every certified method known here. On the weak side, one explicit small-field step (part 1, part 1b) has block Poincaré constant $\frac{4}{9}$, fluctuation size $\frac{3}{2}g$ per link, and a proved propagator decay rate $\kappa \sim 10^{-3}$ per lattice unit, against an expected order-one value that no source proves. Its remainder threshold scales as κ^4; a scaling estimate with crude constants puts it at $g^2 \sim 10^{-12}$ if κ is of order one and at $g^2 \sim 10^{-35}$ at the proved rate. Even the first figure lies **twelve orders of magnitude below the physical onset of the running at** $g^2 \simeq 1$, where the strong side's deficit is a factor 400 in β_W. On that estimate, constant-chasing within this method does not close the weak side, and H1 is a methods problem at order-one coupling.

Division of labour. Hamiltonian methods for T1, T2, the small-volume theorem, the upper bounds, the exact identities and the final gap extraction; the Euclidean polymer expansion for the renormalization steps; and, for the intermediate region, the finite-volume mixing conditions, which make the gap at a given coupling follow from a finite verification and turn the problem into the entry of the trajectory of effective interactions into the open set of completely analytical interactions, a sufficient route whose two-number form is the meeting of the reach g^2_{RG} of the small-field renormalization with the reach g^2_{DS} of the verification (finite-verification note).

What this programme has added is a map with constants: six named statements, two proved, the finiteness clause reduced to a single correlator, seven routes closed with explicit numbers, for the large-field region an action floor and a model mode count with its closure still open and the norm obstruction met by the crude Hamiltonian proof, three places where the non-abelian structure is isolated, the exact identities and sum rule of Section 2, rigorous explicit thresholds on the strong side of the intermediate region and labelled estimates on its weak side. Several of the closures are corrections of claims made earlier in the same series, and each is recorded with the computation that forced it.

7. Where the map ends

The mass gap for $SU(3)$ in four dimensions is not proved here. The strong side of the confinement scale carries rigorous explicit thresholds and the weak side explicit estimates for one step, the two standing reasons for stopping have been researched rather than left as reasons, and what remains can be stated in one sentence.

The two reasons, researched. A Griffiths-type correlation inequality would give monotonicity of the string tension and of decay rates in channels with vanishing mean, and it transfers control from weaker to stronger coupling only; on this map it replaces the three strong-side blocking steps and leaves the weak side untouched, so it is a simplification and not an unblocking. Even the

vacuum-sector gap's monotonicity would not follow, the truncated plaquette correlator having two terms that both increase. A certified verification of the mixing condition at $\beta_W \simeq 6$ is a supremum over boundary conditions of an integral in thousands of dimensions, while certified polymer enumeration converges only where the expansion does, which the explicit criteria here place at $\beta_W \lesssim 10^{-2}$, with no proof that it reaches $\beta_W \simeq 6$; so it cannot be supplied where it is needed, by any method known to this author.

The asymmetry of the two sides. The strong side is proved to $\beta_W = 0.0135$ (Dobrushin, $g^2 > 444$) against a physical crossover at 5.7, a factor of about 400. The weak side, computed explicitly for one blocking step, has its threshold controlled by the fourth power of the fluctuation propagator's decay rate, which no source states explicitly and which two explicit arguments put at 10^{-3} per lattice unit against an expected order-one value. A scaling estimate with crude constants puts the threshold at $g^2 \sim 10^{-12}$ even with an order-one rate, which is assumed, twelve orders of magnitude below the physical onset of the running at $g^2 \simeq 1$, and at $g^2 \sim 10^{-35}$ with the proved rate. On these estimates, sharpening constants within the present methods closes neither side, and the weak side misses by a margin that no sharpening addresses.

What is open. Control of the renormalization steps at couplings where the fluctuation is not small compared with the nonlinearity, with no expansion in any parameter. That is the mass-gap problem for $SU(3)$, stated as precisely as this programme can state it, and it is new mathematics. The small-field steps at weaker coupling also lack explicit constants (§6), a quantitative deficit on the same trajectory.

8. Consequence for STATE

Correction of 2026-10-02. STATE and other notes citing this map should use the labelled thresholds of §6: $g^2 > 444$ (Dobrushin) and $g^2 \geq 1059$ (polymer, all representations) for the $SU(3)$ Wilson transfer matrix; Yarotsky's existential threshold, $g_0^2 \simeq 10^{101}$ when made explicit, for the Kogut–Susskind Hamiltonian; and estimates only on the weak side. This map no longer supports a phase-transition equivalence for T2′, a failure of T2′ for $SU(N \geq 5)$, a closed large-field region, a verbatim Euclidean transfer or a Hamiltonian truncation bound. No STATE or catalog edit is made in this correction.